AWANTIKA SRIVASTAVA

Data Scientist | GenAI & LLM Engineer | ADAS

+91-8920482037 | sawantika81@gmail.com | LINKEDIN | Github

PROFILE SUMMARY

Data Scientist with **2+ years of experience building end-to-end Machine Learning and Generative AI solutions**. Strong expertise in **Large Language Models (LLMs), Retrieval Augmented Generation (RAG), SQL-driven data extraction, and ML pipeline deployment**. Experienced in designing scalable **AI systems from data preparation to model deployment and stakeholder insight generation**. Proven ability to solve complex business problems using **ML, Deep Learning, and data analytics techniques**.

CORE TECHNICAL SKILLS

- **Programming & Data Handling:** Python, Numpy, Pandas, Matplotlib, Scikit-learn, C++, SQL.
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Linear Algebra (Vectors, Matrices)
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, BERT, Vision Transformers (ViT), Self-Attention, Cross-Attention, Multi-Head Attention, Sequence Modeling, TensorFlow, PyTorch, Keras, Time-Series Forecasting, ARIMA, SARIMA
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Feature Extraction, Multi-Modal Fusion Concepts
- **NLP**: Text preprocessing, Tokenization, Chunkings, Transformers, Sentiment Analysis, Semantic Search, Transformer-based models, NLTK, SciPy
- **Generative AI:** LLMs, Retrieval Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search
- **MLOps / LLMOps:** Docker, CI/CD, Model Deployment, Pipeline Automation
- **Tools:** SQL, Pandas, NumPy, Data Extraction Pipelines, Git, Linux, AWS (EC2), PowerBI

EXPERIENCE

Data Scientist | PPS International Pvt. Ltd. **January 2024-Present**

- Built and deployed **end-to-end Machine Learning pipelines** covering data ingestion, preprocessing, feature engineering, model training, evaluation, and deployment.
- Developed scalable **Python-based AI solutions** for real-time and batch inference use cases.
- Designed and implemented **RAG-based LLM pipelines** to retrieve relevant information and generate context-aware responses.
- Utilized **SQL-based data extraction and transformation** to prepare high-quality datasets for model training.
- Applied advanced **machine learning and deep learning algorithms** to solve predictive modeling problems.
- Performed **model debugging, evaluation, and optimization** to improve model accuracy and robustness.
- Implemented **MLOps workflows** including containerization, version control, and automated deployment.
- Collaborated with cross-functional stakeholders to translate **business problems into AI-driven insights and solutions**.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- Designed and **deployed a real-time computer vision**–based ML system to detect unsafe driver behaviors from continuous video streams.
- Trained and **optimized CNN-based** object detection **models (SSD MobileNet** architecture) to perform real-time inference on video data.
- **Implemented end-to-end ML pipelines** covering data ingestion, preprocessing, model training, evaluation, and production inference.
- Achieved **20–25 FPS** real-time processing with **<150 ms** inference latency by optimizing models for **deployment**.
- Built and integrated **a Flask-based** web **interface** to **visualize** detections and automatically recorded 30-second event clips, reducing manual **review** effort.

LLM & RAG-Based AI Chatbot | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | Link -https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology